MIL-C-52961(ME)
2 April 1980

MILITARY SPECIFICATION

CAMERA, COPYING, LITHOGRAPHIC, DARKROOM,

MOBILE, PROCESS

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a mobile process copying camera.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-L-20	- Lacquer, Camouflage.
TT-P-1757	- Primer, Coating, Zinc Chromate, Low Moisture Sensitivity.

FSC 3610

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-V-173	- Varnish, Moisture and Fungus-Resistant (for the Treatment of Communications, Electronic and Associated Electrical Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-C-675	- Coating of Glass Optical Elements (Anti-Reflection).
MIL-T-704	- Treatment and Painting of Materiel.
MIL-F-3541	- Fitting, Lubrication.
MIL-C-3615	- Cameras, Copying, Lithographic, Packaging of.
MIL-G-3859	- Grease Guns, Hand, High Pressure, Lever Operated; Cartridge and Bulk Loading (14 Ounce and 21 Ounce Capacity).
MIL-W-5086	- Wire, Electrical, 600-Volt, Copper, Aircraft.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-C-52286	- Cable and Lead Assemblies, Power, Electrical, 300 to 600 Volts, 5 to 230 Amperes.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-150	- Photographic Lenses.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

DRAWINGS

ME

TA13222E5000	- Camera, Copying, Lithographic, 30 In. Lg. 24 In. Wide, Negative Size, 208V, 3 Phase, 60 Cycle.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL FIRE PROTECTION ASSCCIATION (NFPA)

Std. No. 70 - The National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110).

SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents.

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 South San Pedro Street, Los Angeles, CA 90013).

3. REQUIREMENTS

3.1 Description. The camera shall be in accordance with Top Assembly TA13222E5000, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the camera. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one camera for examination and testing within the time frame specified (see 6.2) to prove prior to stating production that his production methods will produce cameras that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.5).

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.9).

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3.4 Capacities, dimensions, and weight. The camera shall have the following capacities, dimensions, and weight:

Net weight: Maximum	3500 pounds
Operating height: maximum	72 inches
Copy Board:	
Opaque	36 x 48 inches
Transparent	21 x 27 inches
Range of Reproduction:	
19 inch lens	33 to 300 percent
24 inch lens	50 to 200 percent
Vacuum Back:	
Sensitized Material	
Minimum	4 x 5 inches
Maximum	24 x 30 inches

3.5 Rigidity. There shall be no observable shift of the projected image due to any sudden lifting of one corner of the camera skid or movement of personnel about the camera when the camera is mounted in a van-type mode. The skid platform shall be flat and true to line within 1/4-inch over its entire area when supporting the camera assembly, either on its wheels or on an uneven floor.

3.6 Vibration absorption. No movement or vibration of the projected image shall be observable when the vacuum exhausters are operating.

3.7 Shock absorption. When 10 foot-pounds of force are instantaneously applied upon the rear portion of the camera rail, the shock shall be dampened within 5 seconds after impact.

3.8 Accuracy. When supporting copy weighing up to 25 pounds, the camera shall reproduce, within its rated capacity of enlargement and reduction and within its field of lens coverage, all copy in accurate size and proportion to within 0.005 inch of the required dimensions. The photographic planes of the copyboard, transparency holder, lensboard, ground glass, and vacuum back shall be flat and parallel and shall maintain their parallelism throughout their range of movement to within 0.005 inch. This accuracy shall be retained following the cycle of positioning for transit, movement in a van body across rough roads, and re-expansion of the camera into operating postion. Shims may be used to obtain this accuracy.

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3.9 Lead screws and counters. Lead screws for the camera drive controls shall be machined to 0.005 inch tolerance over any increment of their length. Counters shall be coupled to the lead screws and shall be geared to operate safely at speeds of not less than 7,500 figures per minute. Counters shall have visible figures and shall indicate the location of the lensboard and copyboard from the ground glass in increments of 0.01 inch.

3.10 Copyboard assembly. The copyboard and transparency holder assembly shall be pivotally mounted on the copyboard carriage so that both the copyboard and the transparency holder are held within 0.002 inch of the ideal photographic plane when in position for copying. The copyboard shall be equipped with vacuum parts arranged so that numerous sizes of copy can be held centered on the board. The surfaces of the copyboard and transparency holder shall be flat within 0.002 inch. Center lines and centrally located copy size scribe lines shall be permanently scribed on the surface of the copyboard. The center lines of the copyboard and transparency holder in the operating position shall be coincident with the optical axis of the lens. The operating surfaces of the copyboard and transparency holder must be parallel with the vacuum back surface within .005 inch. The copyboard shall accept and hold in a flat plane any size of copy prepared on paper, sheet metal, or 3/8 inch plywood up to and including 36 by 48 inches. The transparency holder shall accept transparencies up to and including 21 x 27 inches.

3.11 Copy board illumination. The copy board shall be illuminated by four solid state, pulsed xenon lamps. The lamps shall operate on a 208-volt, lamp controller and shall be equipped with an automatic voltage regulator.

3.12 Bellows. The bellows shall be of rubberized duck material to form a single accordion-type bellows unit capable of contracting and extending with the full range of travel of the lens-board carriage.

3.13 Darkroom connecting frame and light barrier. The darkroom connecting frame shall provide a flexible lighttight joint between the camera back and the darkroom wall opening. The sliding curtains, flexible joint, the entire camera back, bellows fittings, and lensboard assembly shall provide a lighttight barrier between the camera room and darkroom.

3.14 Lighttightness. All openings in the camera back shall be provided with a light seal to insure that the entire camera back will be lighttight with no evidence of light leaks when viewed from the darkroom. Light seals shall be of a flexible material which will not transmit vibration to the camera.

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3.15 Vacuum back. The perforated vacuum back surface shall have concentric rectangles permanently marked and clearly visible under darkroom illumination. The rectangles shall be centered about the optical axis and shall correspond in size to commercial standard film dimensions. The lines shall be fused into or absorbed by the surface so as to cause no surface distortion of the film. The holding face shall be plane and true to within 0.002 inch and shall have no surface irregularities of a size that will result in local distortion of the film negative. The porosity of the vacuum back and the suction of the pump shall be sufficient to hold any size of film up to the rated capacity of the camera, through zone control only, without the aid of additional mechanical devices. The emulsion surface of the film mounted on the vacuum back shall occupy to within 0.002 inch the same photographic plane of the front surface of the ground glass when the vacuum back door is locked into exposure position.

3.16 Ground glass. The ground glass shall be pivotally mounted so that it may be positioned directly in line with the lens for alinement and focus of the copy image or swing out of the way of the vacuum board. The front surface must be flat within .002 inch. When in position for alinement and focus, it must be parallel with the copy board within .005 inch, and be located within .002 inch of the focal plane of the vacuum back. The grained surface of the ground glass shall resolve not less than 20 lines per millimeter in a projected image. A series of rectangular shaped grid lines shall be scribed in the surface and accurately oriented about the center point of the glass.

3.17 Focusing system. The camera shall be calibrated to a predetermined focusing system for positioning the lens board and copyboard by charted readings in 1-percent increments for any ratio of enlargement or reduction over the specified ranges of the lenses furnished. Such positioning shall provide the maximum resolving power for the lenses, as well as accuracy of both horizontal and vertical image dimensions on the ground glass, across the maximum rated field of coverage for all ratios of reproduction.

3.18 Focusing charts. Three copies of a focusing chart shall be provided for each lens with legible counterreadings for each percentage of reproduction and enlargement over the specified range of reproduction (see 3.4). The focusing charts shall include the following information: Lens make, serial number, nominal focal length, nominal maximum aperture, back focal distance, equivalent focal length, nodal point separation, lens thickness, and camera serial number. The charts shall be approximately 9 by 12 inches and shall be laminated in transparent plastic.

3.19 Test chart. A metal test chart for checking the calibration and accuracy of the camera shall be provided in a protective container. The test chart shall consist of a hard aluminum alloy sheet not less than 0.032

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inch thick on which a white lacquered or enameled surface forms the base for a photographically transferred image from the master grid used in preparing the ground glass. The grid lines shall be spaced at right angles within individual and overall tolerances of 0.003 inch. The image shall be given a hard finish by heat treatment or other protective means. The image on the test chart shall be made so that conjugate lines and numbers will match when the test traget is placed on the copyboard and viewed through the ground glass at a ratio of 1:1.

3.20 Lenses. Two interchangeable, color-corrected, process lenses conforming to MIL-STD-150, Type IV, shall be provided with each camera. The nominal focal length of the lenses shall be 19 inches and 24 inches respectively, and the relative aperture shall be f/11. The lenses may have a larger relative aperture; however, the range shall be between f/11 and f/64. The maximum variation of focal lengths when measured photographically through Wratten A, B, or C5 filters over the visible and near ultraviolet spectral range shall be not more than 0.2 percent. The half-angular field of coverage of the lenses shall be not less than 22 degrees for both lenses. Each lens shall be mounted inside a barrel which is mounted on a lens-mount plate conforming to the drawings. Aperture markings shall be visible when observed from the right side of the camera while facing the copyboard. Each lens shall be marked with the manufacturer's name, lens name, serial number, nominal focal length in inches, and maximum aperture. Lenses shall withstand 155° F ± 2° F for not less than four hours and minus 65° F ± 2° F for not less than twelve hours without evidence of damage such as cracking, chipping, breaking.

3.21 Lens coating. All air-glass surfaces of each lens shall be coated with a reflection-reducing coating conforming to MIL-C-675.

3.22 Lens distortion. The distortion in either lens at f/11 aperture shall not exceed that which would cause a point in the image plane to be displaced from its theoretically distortion-free position by more than 0.05 mm at a ratio of 1:1 reproduction throughout an angular field of 0 to 22 degrees.

3.23 Diaphragm aperture control. Provisions shall be incorporated to correlate the f-stop openings to the ratio of reproduction for equivalent light transmission of the lenses throughout the range of reproduction specified in 3.4. The iris diaphragm shall be graduated and marked as follows: f/11, f/16, f/22, f/32, f/45, and f/64. It shall be capable of incremental intermediate stops between the major stops.

3.24 Resolving power. The resolving power of the lenses in terms of lines per millimeter (mm) in the image plane shall be not less than the values shown in Table I.

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Table I. Resolving Power

Angle from center (Degrees)	24-inch lens (Lines per mm)	19-inch lens (Lines per mm)
0	23	25
5	20	25
11	20	25
16	12	22
22	12	4

3.25 Lens accessories. The following set of accessories shall be provided for each lens; a lens cap; a lens box; seven interchangeable filter frames; a dusttight filter case with slots to accommodate the filter frames; and a guide detailing use and care of the lenses. Both sets of accessories shall be enclosed in a felt-lined protective compartmentized box having a hinged cover and lock.

3.26 Electrical requirements. The camera shall operate on 208V, 3 phase, 60 Hertz alternating current electrical supply. The electrical equipment shall be as shown on the drawings. Motor receptacle connector, cable assembly, and dummy connector shall conform to applicable portions of MIL-C-52286, and Figures 1 through 6 therein. The electrical wire shall conform to MIL-W-5086. All electrical equipment and wiring shall conform to NFPA No. 70. Helical, spring-wound, rubber-covered cord shall be used from hinged duct to camera lensboard and copyboard. All wiring shall be color coded and circuits numbered. A plastic-enclosed or an etched metal wiring diagram detailing color coded wires and identifying switches and controls by name and part number shall be mounted on an electrical box or other suitable location.

3.27 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.27.1 Lubricants. The camera shall operate as specified herein when lubricated with military lubricants (see 6.4).

3.27.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859, Type I. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand operable, quick-disconnect fasteners may be removed to provide accessibility.

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3.27.3 Delivery lubrication service. When delivered, all components requiring lubrication, except those for which preservatives are specified in MIL-C-3615, shall be lubricated. Grease lubrication, including lubrication of sealed bearings, shall be with standard military thermally stable greases conforming to MIL-G-10924 or MIL-G-23827, as applicable. Each item of equipment shall be tagged in a conspicuous place to indicate the temperature range and grades of greases used.

3.28 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173; Composition I or II as applicable (see 3.28.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.28.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 102, Southern California Air Pollution Control District.

3.30 Instruction plates. Each camera shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the camera. Instruction plates shall conform to MIL-P-514, Type III.

3.31 Treatment and painting. The portions of the camera and its components normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704. All parts of the camera which may reflect light into the lens or onto the film shall be cleaned and treated in accordance with MIL-T-704, given a primer coat conforming to TT-P-1757 and a finish coat conforming to TT-L-20, color black.

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3.32 Storage. Camera and all ancillary equipment and spare parts shall withstand prolonged storage at temperatures of minus 65° F to 155° F and relative humidities ranging from 10 percent to 90 percent without degrading operation of camera when returned to normal operating conditions (see 4.5.2.9).

3.33 Government-loaned property. Unless otherwise specified (see 6.2), map reproduction van body, semi-trailer mounted, will be loaned by the Government (see 6.3).

3.34 Safety.

3.34.1 Exposed parts. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded. All fragile components, such as transparency holder glass, ground glass, zenon lights, and flash lamp shall be mounted and guarded to prevent breakage and damage during use and transit. Protective devices shall not impair the operating functions.

3.34.2 Operations. All operations of the camera are to be designed to be safe so as to avoid hazards to any operating or transient personnel in the vicinity of the camera.

3.35 Human engineering. The design of the camera shall allow for operation by personnel, both male and female, from 5th to 95th percentile as defined in MIL-STD-1472.

3.36 Workmanship. All parts, components, and assemblies of the camera including forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, pits, scale, and other harmful extraneous material. All exposed edges which could cause injury to an operator shall be rounded or chamfered.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent

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necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model. Reassembly with replacement parts and retesting shall be the responsibility of the contractor. Government-loaned property shall be excluded from the requirements of this paragraph.

4.2 Classification of inspections. The inspection requirements specified herein are as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction camera shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. The preproduction camera shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection or for performing the inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Examination. Each camera shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each camera shall be tested as specified in 4.5.2.1 and 4.5.2.3 through 4.5.2.7. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The cameras shall be examined as specified herein for the following defects:

101. Cable assembly not as specified.
102. Electrical equipment not operating.
103. Hinges, doors, or counterbalanced parts fail to operate.
104. Manual drive handles or other operating controls bind.
105. Stops or guides inadequate or missing, or screws bind when the camera bed assembly is raised or lowered to and from the operating position.

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106. Center skid wheels not 1/4 inch lower than end wheels when lowered into operating position.
107. Skid-wheel handles do not operate.
108. Zone control does not control the suction area of the vacuum back as indicated by the zone graduations.
109. Solenoid-controlled shutter not operable from the camera room or the darkroom.
110. Aperture and lens markings not as specified.
111. Suction of the transparency holder and zoning of the copyboard not controllable by the knobs provided.
112. Metal cover frame, transparent film sheets, and clips or clamps not furnished with the transparency holder.
113. Flash lamp cannot be brought in front of lens at all positions of lens, or does not contain a 60-watt bulb.
114. Capacities, dimensions, and weight not as specified.
115. Nonconformance to drawings.
116. Skid platform not as specified.
117. Lens accessory box improperly assembled or not mounted securely to lens-board carriage.
118. All accessories not protected from damage in lens accessory box as specified.
119. Counter readings of focusing charts not as specified.
120. Fungus and moisture resistance not as specified.
121. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Prior to test, those parts of the camera normally requiring lubrication shall be lubricated with oils and greases specified in 3.27.

4.5.2.2 Rigidity, vibration absorption, and shock absorption. The camera shall be mounted in the Government-loaned 2-1/2 ton truck (see 3.33). Printed copy with halftone dots shall be placed on the copyboard of the camera, and the projected image of the halftone dots shall be observed on the ground glass with a 40-power microscope.

4.5.2.2.1 Rigidity. The projected image shall be observed with the microscope while two or more individuals walk about the van. One corner of the camera shall then be lifted 1/2 inch above the floor while the projected image is being viewed on the ground glass. Any observable image shift shall constitute failure of this test.

4.5.2.2.2 Vibration absorption. With the vacuum blowers in operation, the projected image shall be observed with the microscope. Any observable image shift shall constitute failure of this test.

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4.5.2.2.3 Shock absorption. A 10-pound lead weight shall be dropped from a height of 12 inches upon the rear portion of the camera rail while the projected image is being observed on the ground glass. Incomplete dampening within 5 seconds after impact shall constitute failure of this test.

4.5.2.3 Accuracy. The camera shall be tested optically and mechanically for accuracy. The optical test shall utilize the focusing system, the test chart, and the ground glass. Tests shall be conducted on a minimum of 5 ratios of enlargements and reduction on each lens, and shall include the ratio 1:1 and the maximum and minimum positions covered by each lens. The projected image shall be compared to the size of the test chart to determine the size of enlargement or reduction. Mechanical tests shall be conducted with a precision trammel equipped with a dial indicator. Tests shall include measurements to determine parallelism of planes and the ability of the camera controls to position the lens board and the copyboard for any desired enlargement or reduction and to repeat these positions. During this test, the location and functioning of the limit switches and stops on the camera ways and carriages shall be tested to insure that they are properly positioned to prevent excessive movement or interference between the carriages when the camera is power driven or hand operated. Nonconformance to 3.8, 3.9, 3.10, 3.15, 3.16, and 3.17, or evidence of excessive movement or interference between the carriages when the camera is power driven or hand operated shall constitute failure of this test.

4.5.2.4 Resolving power. Each lens shall be tested individually at a magnification of 1:1, at an aperture setting of f/22. The resolving power tests shall be conducted using targets and target layout in accordance with MIL-STD-150. Test targets shall be of high-contrast type with dark lines on a light background; illumination shall be by solid state, pulsed xenon camera lights with a total power of 6 kW. Nonconformance to 3.24 shall constitute failure of this test.

4.5.2.5 Lens distortion. The lens distortion shall be tested visually in accordance with MIL-STD-150, Method 30, with the lens mounted on the camera at a reproduction ratio of 1:1. The test shall be made with the lens at maximum aperture and the light source shall be the solid state, pulsed xenon camera lamps with a total power of 6 kW. A point in the image plane that is displaced from its theoretically distortion-free position by more than 0.05 mm at a reproduction ratio of 1:1 throughout an angular field of 0 to 22 degrees shall constiture failure of this test.

4.5.2.6 Lighttightness. The camera shall be set up for operation by positioning it in a wall opening of the required size with the rear of the camera in the darkroom. The bellows shall be extended full length and the shutter shall be closed. All light in the darkroom shall be extinguished, and the solid state, pulsed xenon lamps in the camera room shall be directed

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toward the copyboard. The inspector shall remain in the darkened darkroom not less than 5 minutes prior to conducting the test to allow the inspector's eyes to adjust to the darkness. The darkroom wall and bellows shall then be observed for light leaks. Evidence of any light leaks shall constitute failure of this test.

4.5.2.7 Vacuum system. The vacuum back, vacuum copyboard, and transparency holder shall be tested for adherence by using a range of film sizes from 4 by 5 inch to 24 by 30 inches on the vacuum back and transparency holder, and copy sizes from 9 by 9 inches to 36 by 48 inches on the vacuum copyboard to assure satisfactory adherence. The film to be used for this test shall be the same as used for the resolution test in 4.5.2.4. Film may be used to represent copy sizes from 9 by 9 inches to 24 by 30 inches. The maximum copy sheet size shall be tested with a 36 by 48 by 3/8 inch plyboard. Lack of adherence of any sheet or board shall constitute failure of this test.

4.5.2.8 Vibration resistance. The camera shall be mounted in the Government-loaned semi-trailer mounted van body and shall be positioned for transit. The complete vehicle with the camera mounted therein shall be driven not less than 20 miles, of which not less than 10 miles shall be unsurfaced, stabilized, or partially surfaced earth or gravel roads. The test shall include ten abrupt starts, ten abrupt stops, and five sharp turns at maximum safe speeds in each direction (right and left). The route shall be approved by the contracting officer. At the completion of the test run the camera shall be operated. Any evidence of damage to the camera or nonconformance to 3.8, 3.9 3.10, 3.15, 3.16, and 3.17 shall constitute failure of this test.

4.5.2.9 Storage. The camera and all ancillary equipment and tools shall be placed in a temperature test chamber at normal room temperature. The temperature of the chamber shall then be raised to plus 155° F ± 2° F and maintained at this temperature for not less than 4 hours. The temperature shall then be returned to normal room temperature and the camera and equipment shall be examined for damage. The temperature in the chamber shall then be lowered to minus 65° F ± 2° F and maintained for not less than 12 hours. The chamber shall then be returned to normal room temperature, and the camera and equipment shall be examined for damage and operated to verify the performance in accordance with 4.5.2.3, 4.5.2.4, 4.5.2.5, 4.5.2.6, and 4.5.2.7; nonconformance in any of the above and nonconformance to 3.20 shall constitute failure of this test.

4.5.2.10 Human engineering. The human engineering requirements shall be evaluated during testing. Nonconformance to 3.35 shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packing, and marking for Level A or B shall be examined and tested in accordance with the Quality Assurance Provisions of MIL-C-3615. The preservation, packing, and marking for commercial packaging shall be exaamined to determine compliance with MIL-STD-1188.

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5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B, or Commercial as specified (see 6.2).

5.1.1 Level A. Each complete camera shall be preserved in accordance with the Level A preservation and packaging requirements of MIL-C-3615.

5.1.2 Level B. Each complete camera shall be preserved in accordance with the Level B preservation and packaging requirements of MIL-C-3615.

5.1.3 Commercial. Each complete camera shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A. Each complete camera, preserved as specified in 5.1, shall be packed in accordance with the Level A packing requirements of MIL-C-3615.

5.2.2 Level B. Each complete camera, preserved as specified in 5.1, shall be packed in accordance with the Level B packing requirements of MIL-C-3615.

5.2.3 Commercial. Each complete camera, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The camera is intended to be used in mobile vans for the precision copying of maps and other printed matter for photolithographic purposes.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction model (see 3.2).

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(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 Government-loaned property. The contracting officer should arrange to lend the property specified in 3.33.

6.4 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the camera covered by this specification as contained in Federal Supply Catalogue, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.5 Preproduction. Any changes or deviations of production cameras from the approved preproduction model during production will be subject to the prior approval of the contracting officer. Approval of the reproduction model will not relieve the contractor of his obligation to furnish cameras conforming to this specification.

6.6 Replacement parts and tools. The following replacement parts and tools should be furnished with the camera:

(a) One ground glass (focusing).
(b) One ground glass for transparency holder.
(c) One set of trammels with dial indicator.
(d) Four xenon bulbs.
(e) Oiler, hand, 1/2 pint capacity, NSN 4930-00-537-8977 - 1 ea.

6.7 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each camera.

6.8 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools) and instructions regarding shipment of camera.

6.9 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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Army - ME

Review activity:
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Preparing activity:
Army - ME

Project 3610-A304

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: MIL-C-52961(ME) Camera, Copying, Lithographic, Darkroom, Mobile, Process

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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